

WEEK 3 REPORT

Financial Data Analysis Using Power BI

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1. Problem Statement

Financial stock data contains information about stock prices and trading activity over different dates. The objective of this project is to analyze Shopify stock performance over time using Power BI and present the results through interactive visualizations.

2. Objectives

The main objectives of this project are:

- To import and prepare Shopify stock data using Power BI.
- To clean and transform the dataset using Power Query.
- To create a dedicated Date Table for time-based analysis.
- To create a Year–Quarter–Month–Day date hierarchy.
- To establish relationships between the Date Table and stock data.
- To create DAX measures for financial analysis.
- To apply time-intelligence calculations.
- To develop an interactive dashboard for stock performance analysis.

3. Dataset

The Shopify Stock Data contains historical stock price and trading information.

Column	Description
Date	Stock trading date
Open	Opening stock price
High	Highest stock price during the day
Low	Lowest stock price during the day
Close	Closing stock price
Adj Close	Adjusted closing stock price
Volume	Number of shares traded

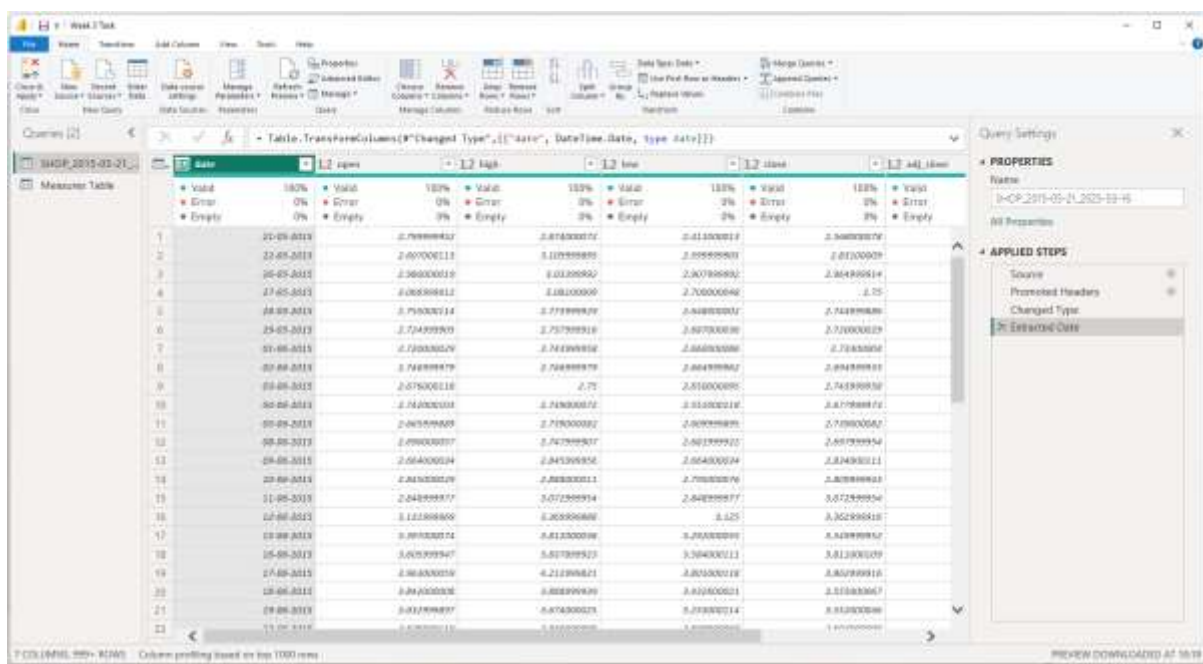
4. Power Query

The Shopify stock dataset was imported into Microsoft Power BI and prepared using Power Query.

The data was checked for appropriate data types:

- Date → Date
- Open → Decimal Number
- High → Decimal Number
- Low → Decimal Number
- Close → Decimal Number
- Volume → Whole Number

Power Query was used to ensure that the dataset was properly prepared for analysis.



	Date	Open	High	Low	Close	Volume
1	2019-01-01	2.78888888	2.81400000	2.78888888	2.81400000	2.81400000
2	2019-01-02	2.80700000	2.81000000	2.80700000	2.81000000	2.81000000
3	2019-01-03	2.80800000	2.81000000	2.80700000	2.81000000	2.81000000
4	2019-01-04	2.80800000	2.81000000	2.80700000	2.81000000	2.81000000
5	2019-01-05	2.79000000	2.79000000	2.79000000	2.79000000	2.79000000
6	2019-01-06	2.79000000	2.79000000	2.79000000	2.79000000	2.79000000
7	2019-01-07	2.79000000	2.79000000	2.79000000	2.79000000	2.79000000
8	2019-01-08	2.79000000	2.79000000	2.79000000	2.79000000	2.79000000
9	2019-01-09	2.79000000	2.79000000	2.79000000	2.79000000	2.79000000
10	2019-01-10	2.79000000	2.79000000	2.79000000	2.79000000	2.79000000
11	2019-01-11	2.79000000	2.79000000	2.79000000	2.79000000	2.79000000
12	2019-01-12	2.79000000	2.79000000	2.79000000	2.79000000	2.79000000
13	2019-01-13	2.79000000	2.79000000	2.79000000	2.79000000	2.79000000
14	2019-01-14	2.79000000	2.79000000	2.79000000	2.79000000	2.79000000
15	2019-01-15	2.79000000	2.79000000	2.79000000	2.79000000	2.79000000
16	2019-01-16	2.79000000	2.79000000	2.79000000	2.79000000	2.79000000
17	2019-01-17	2.79000000	2.79000000	2.79000000	2.79000000	2.79000000
18	2019-01-18	2.79000000	2.79000000	2.79000000	2.79000000	2.79000000
19	2019-01-19	2.79000000	2.79000000	2.79000000	2.79000000	2.79000000
20	2019-01-20	2.79000000	2.79000000	2.79000000	2.79000000	2.79000000
21	2019-01-21	2.79000000	2.79000000	2.79000000	2.79000000	2.79000000
22	2019-01-22	2.79000000	2.79000000	2.79000000	2.79000000	2.79000000
23	2019-01-23	2.79000000	2.79000000	2.79000000	2.79000000	2.79000000

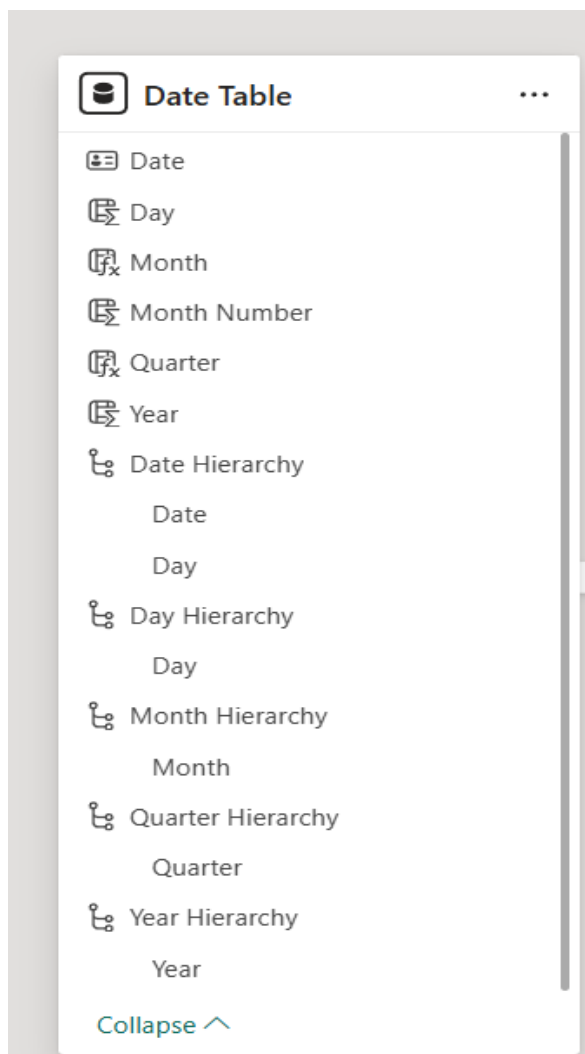
5. Date Table

A dedicated Date Table was created using DAX to support time-based analysis.

The Date Table contains:

- Date
- Year
- Quarter
- Month
- Month Number
- Day

The Date Table provides a structured calendar for performing time-based calculations.



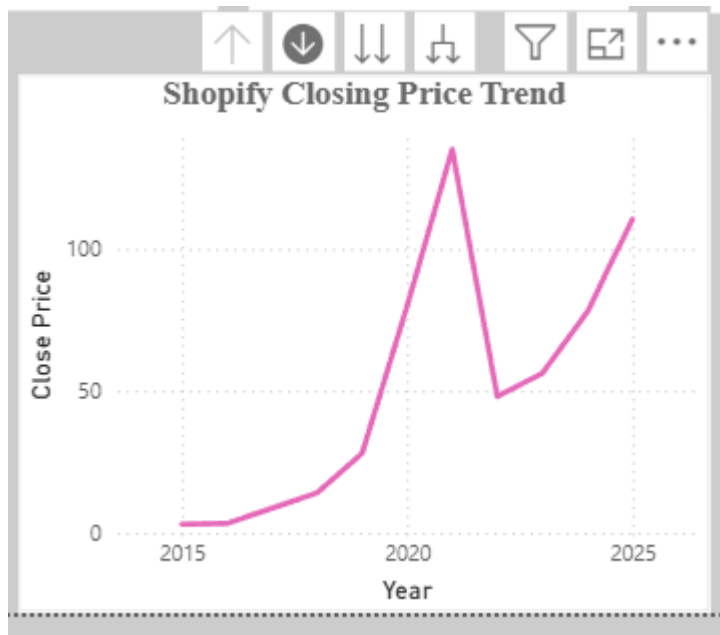
6. Date Hierarchy

A date hierarchy was created to enable drill-down analysis.

Hierarchy

Year → Quarter → Month → Day

This hierarchy allows the stock data to be analyzed at different levels of time, from yearly analysis to daily details.



7. Relationship

A relationship was established between the Date Table and the Shopify Stock Data table.

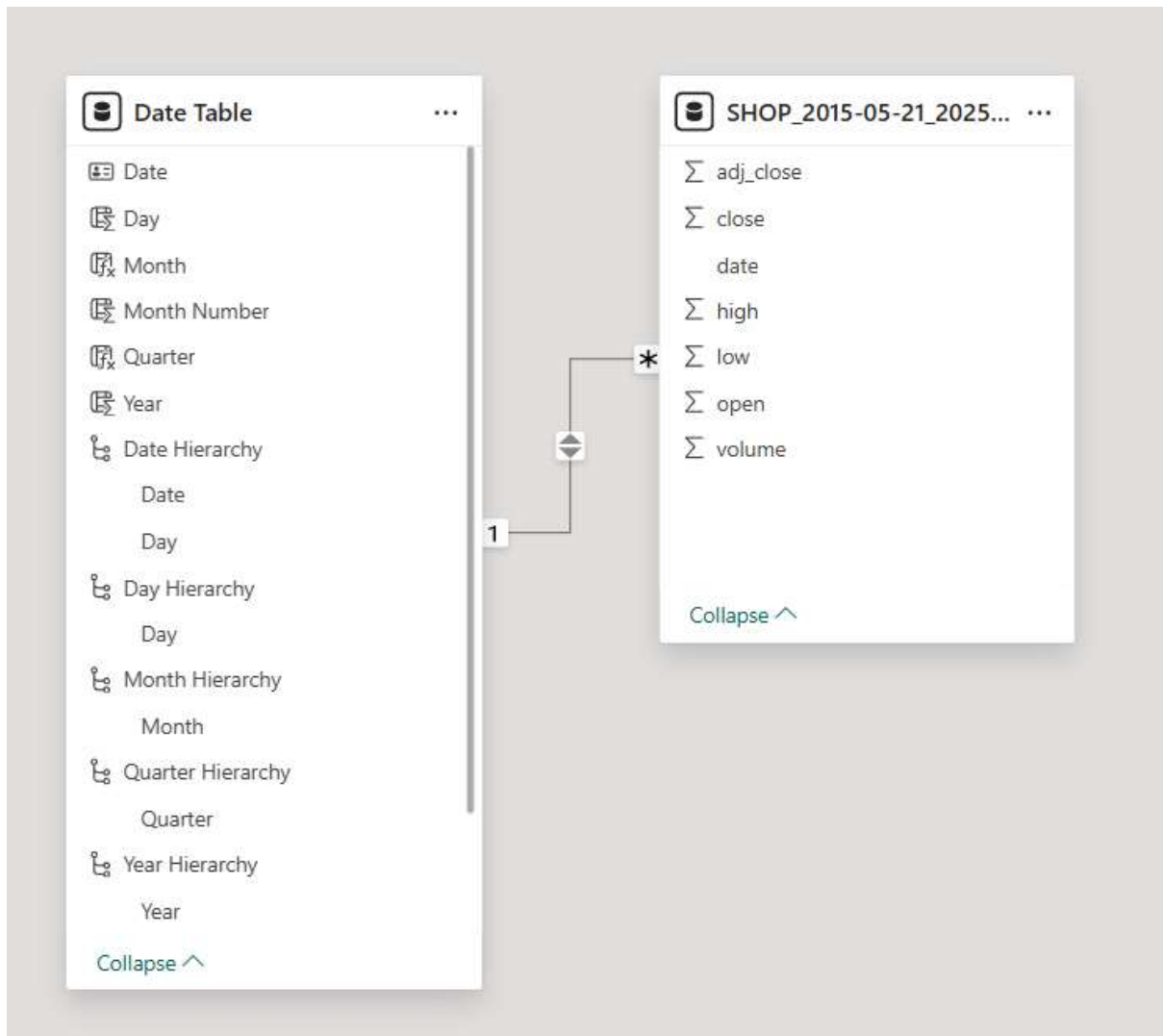
Relationship:

Date Table[Date] → Shopify Stock Data[Date]

Relationship Type:

One-to-Many (1:*)

The Date Table acts as the one-side, while the Shopify Stock Data table acts as the many-side. This relationship allows the Date Table to filter the stock data correctly.

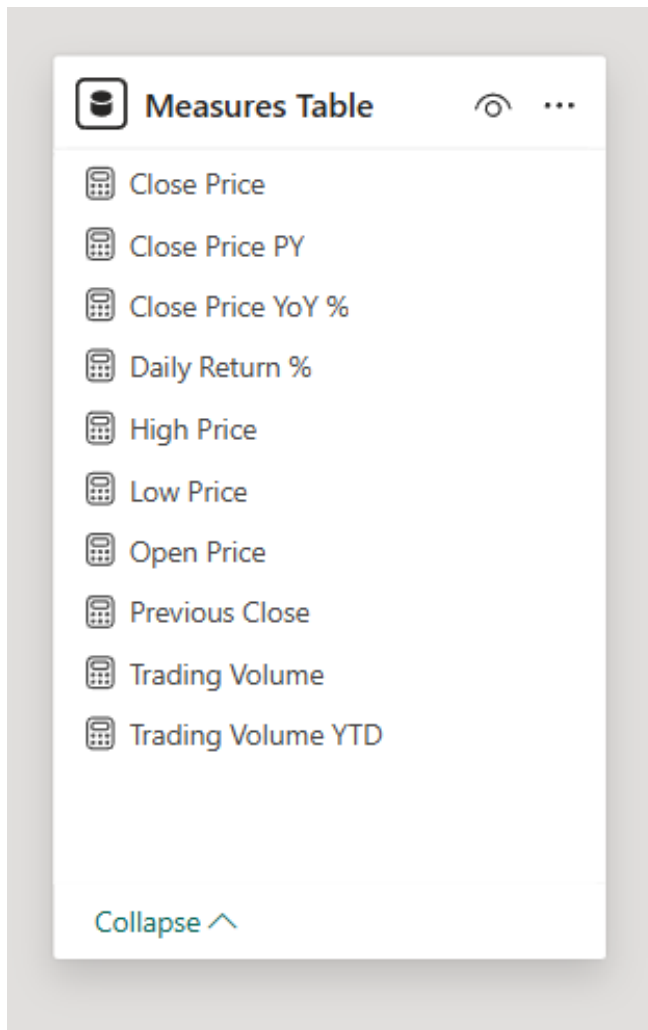


8. DAX Measures

DAX measures were created to calculate important financial metrics.

Measure	Purpose
Open Price	Calculates the opening stock price
Close Price	Calculates the closing stock price
High Price	Identifies the highest stock price
Low Price	Identifies the lowest stock price
Trading Volume	Calculates total trading volume

Measure	Purpose
Previous Close	Retrieves the previous closing price
Daily Return %	Calculates the percentage change in price
Close Price PY	Calculates the previous-year closing price
Close Price YoY %	Calculates year-over-year percentage change
Trading Volume YTD	Calculates year-to-date trading volume



DAX Measures and Time-Intelligence Calculations

9. Time Intelligence

Time Intelligence is used to analyze financial data across different time periods.

In this project, time-intelligence calculations were used to compare stock performance across periods and calculate year-to-date values.

Functions Used

Function	Purpose
DATEADD	Moves the date context to another time period
TOTALYTD	Calculates the value from the beginning of the year up to the selected date
SAMEPERIODLASTYEAR	Compares the selected period with the same period of the previous year

These calculations help in understanding changes in Shopify stock prices and trading activity over time.

10. Dashboard

An interactive **Shopify Stock Performance Dashboard** was created using Power BI.

KPI Cards

- Open Price
- Close Price
- High Price
- Low Price
- Trading Volume

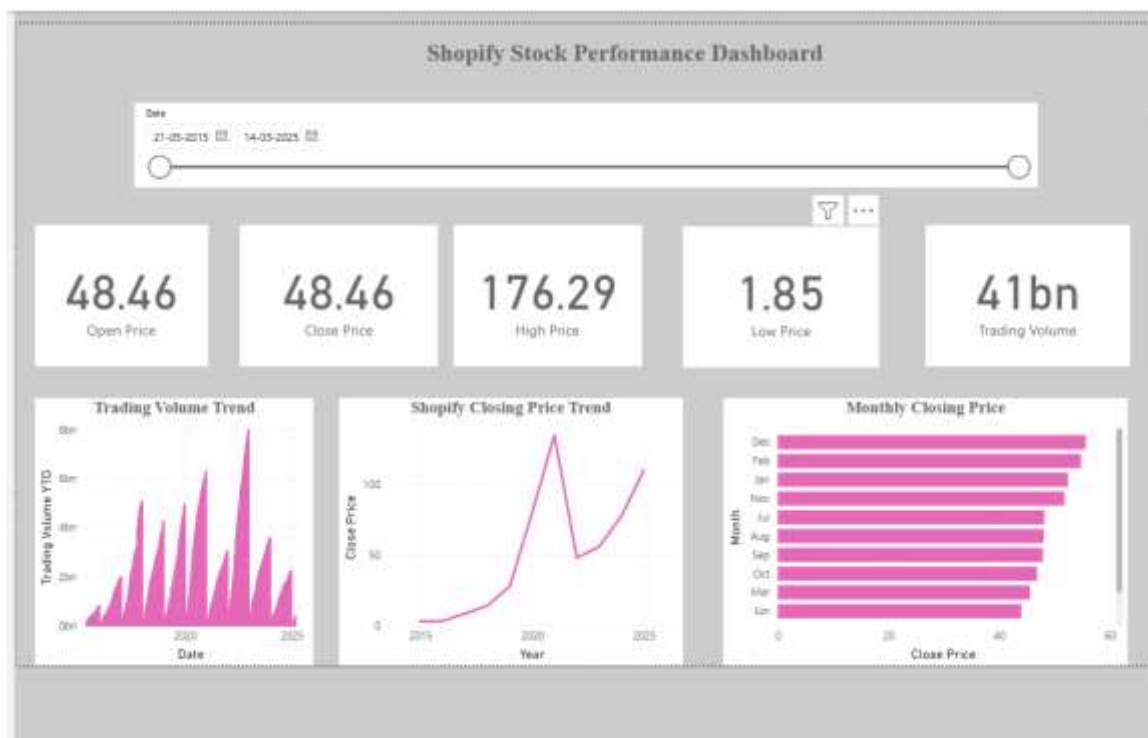
Slicer

- Date Slicer

Visualizations

- Stock Closing Price Trend – Line Chart
- Monthly Closing Price – Bar/Column Chart
- Trading Volume Trend – Line Chart

The Date Slicer allows users to select a specific date range and interactively filter the dashboard.



11. Insights

The dashboard provides the following analytical insights:

- KPI cards provide a quick summary of important stock price and trading metrics.
- The closing price trend shows stock price movement over time.
- The monthly closing price chart allows comparison across different months.
- The trading volume trend shows changes in trading activity over time.
- The Date Slicer allows users to analyze a selected time period.
- Time-intelligence measures support comparison between different periods.

12. Conclusion

The Week 3 project successfully demonstrated the process of importing and transforming financial data, creating a Date Table and date hierarchy, establishing data relationships, developing DAX measures and applying time-intelligence calculations using Microsoft Power BI.

An interactive dashboard was created to provide a clear and structured analysis of Shopify stock prices and trading activity over time.